

PATH FINDING **USING CNN &** **REINFORCEMENT LEARNING**



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Introduction

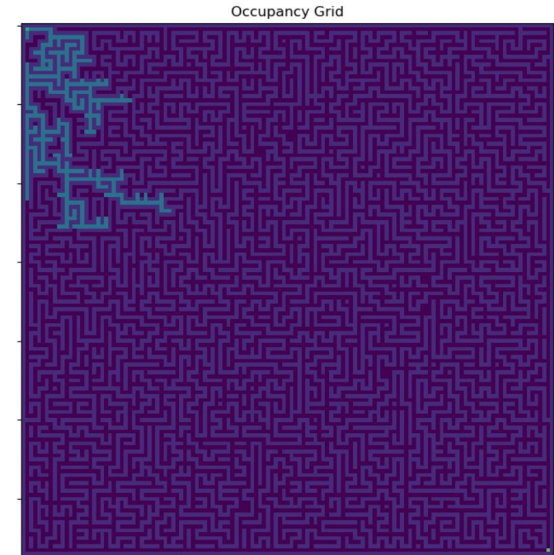
- CNN and Reinforcement Learning - Two most common algorithms in AI
- Path Planning - hot topic
 - Used in Autonomous Driving
- Path Planning
 - Mazes - Obstacles and Free Path
 - Occupancy Grid
 - City Maps

Related work

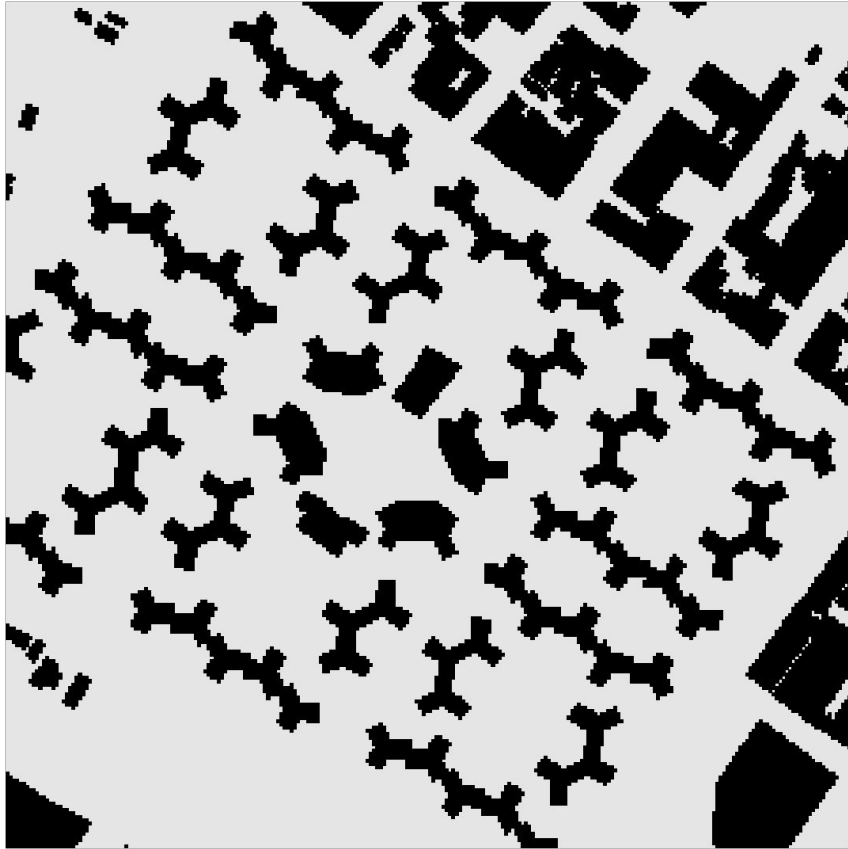
- CNN for Motion Planning
 - Convoluted layers (varying)- 2 Dense layer (Output Layer-> Directions)
 - Used for motion planning
 - Used for image classification
 - Some papers even used just neural network (very old)
 - Many papers used Genetic algorithms
- Reinforcement Learning
 - Dynamic path planning
 - Q Learning - update and behavioural policy
- Using CNN with Reinforcement Learning
 - Very efficient

MAZE

- Fully Observable, Deterministic Environment
- Maze 11*11
- Maze 135*135 ($\frac{1}{8}$ th of the city map dimension)
- City Map Converted To Maze/ Occupancy Grid



New York City Map



Occupancy Grid



CNN (CONVOLUTIONAL NEURAL NETWORK)

- Supervised
- Mazes Generated using Random Maze Generator function from Maze-Lab
- I/P -Matrix of 1's & 0's representing obstacles and free path
- O/P - The directions moved in at each location forming shortest path
 - Shortest Path found using A star
 - O/P ACTIONS = ['^ ', '< ', 'v ', '> '] => [1,2,3,4] and 0 => No action
 - Start and end position =5
 - Training data obtained using A*
- Architecture of CNN dependent on the size of image/matrix
- 2 different CNNs are trained for 11*11 and 135*135
 - 200 SAMPLES
 - 150 TRAINING, 25 VALIDATION, 25 TESTING
 - 135*135 tested on City Map

CNN FOR 11*11 MAZE

- 2 LAYERS OF CONVOLUTION
- EACH WITH MAX-POOLING AND DROP-OUT
- RELU ACTIVATION FUNCTION
- CATEGORICAL CROSS ENTROPY LOSS FUNCTION
- 2 DEEP LAYERS (242 and 121)

```
model.add(Conv2D(32, kernel_size=(3, 3),activation='relu',input_shape=(11,11,1)))  
model.add(MaxPooling2D(pool_size=(2, 2)))  
model.add(Dropout(0.65))  
model.add(Conv2D(64, (3, 3), activation='relu'))  
model.add(MaxPooling2D(pool_size=(2, 2)))  
model.add(Dropout(0.5))
```

CNN FOR 135*135 MAZE

- 1 LAYER OF CONVOLUTION
- MAX-POOLING AND DROP-OUT
- RELU ACTIVATION FUNCTION
- CATEGORICAL CROSS ENTROPY LOSS FUNCTION
- 2 DEEP LAYERS (18225 and 18225)
- Very high accuracy

Worked with city map converted to Occupancy Grid

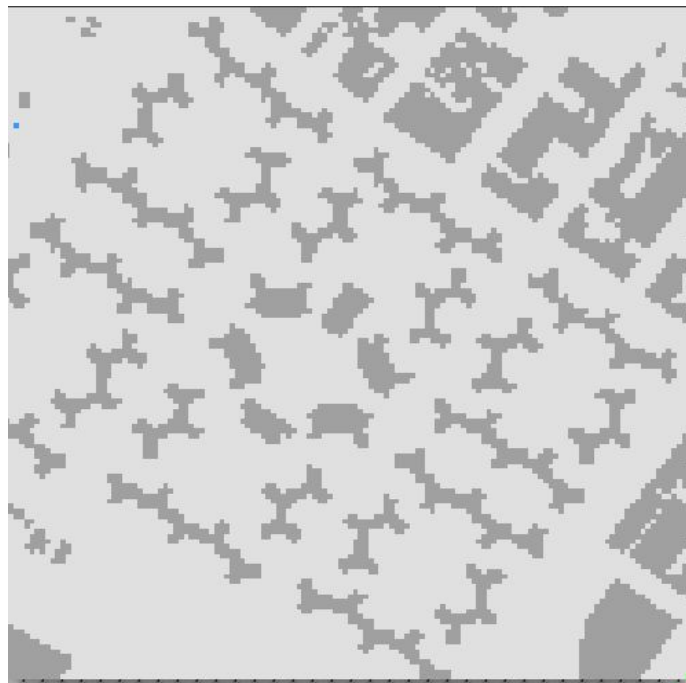
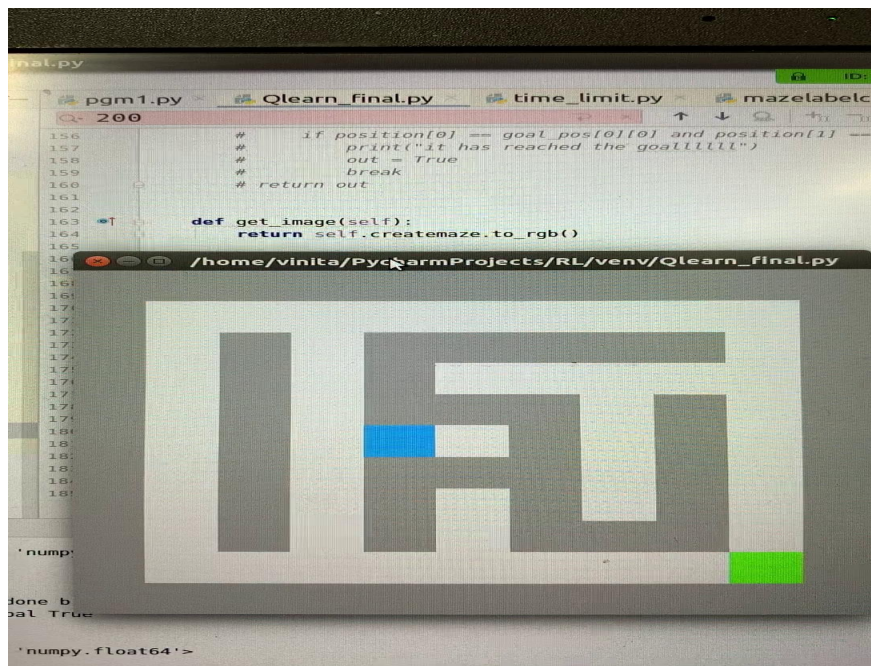
Reinforcement Learning

- Q Learning
 - Agent learns an environment without any prior knowledge.
 - Markov Model: Future state depends only on the current state
 - Off Policy Algorithm : Updation (agent learns optimal policy) and Behavioural policies (agent behaves)
 - State -Action Value pair
- State
 - Path
 - Obstacle
 - Start State
 - Goal State
- Action: (Von Neumann Motion)
 - North , South , West, East
 - Epsilon Strategy Method

Reinforcement Learning

- Rewards:
 - Unnormalized Reward
 - Goal :100, Obstacle: -1, Path : 1
- Learning Rate (α) $0 < \alpha < 1$
 - More exploitation in the beginning and More exploration towards the end.
- Discount Factor (γ) $0 < \gamma < 1$
 - Importance given to immediate of future rewards
 - Exploration strategy: Epsilon greedy strategy (Behavioural policy)
 - Epsilon increases - Random selection; Epsilon decreases- Maximum reward action
- Bellman Equation: (Updation Policy - Agent learns the optimal)

$$Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) + \alpha [r_t + \gamma \max_a Q_t(s_{t+1}, a) - Q_t(s_t, a_t)]$$



RESULTS

Method	CNN	Reinforcement Learning
Accuracy	100%	70%
Time Consumed Training	Very long (couple of hours, depending on the size of i/p)	No need for training
Time Consumed Testing	Very little (less than a few seconds almost 0.2 s)	Very long but converges for small mazes
Advantages	Fast and Efficient for particular cases	Trained dataset not required Generalizes well
Disadvantages	Not Versatile Need Labelled training data Computationally Exhaustive	Gets stuck in the local optima

CONCLUSION AND FUTURE SCOPE

- Train CNN on larger data sets
 - More and less dense mazes together
 - Varying start and end points
- Train CNN on the maps itself
- Improve Reinforcement Learning by
 - fine tuning the hyper parameters such as learning rate
 - Exploration strategy
- Combine CNN and Reinforcement learning to combine the advantages of both the models
- Make the model versatile and fast

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